Yarmouk University
Hijjawi Faculty for Engineering Technology
Communication Engineering Department

Graduation Project I report

Smart Library Monitoring and Predictive Analytics System

Done by:

Sara Ibrahim Sleit

Ansam Ali Alawneh

Supervised by:

Dr. Ayman Jaradat

Dr.Sharief Abdel- Razeq

Semester:

Summer semester

Date:

11/10/2026

Student's Property Right Declaration and Anti-Plagiarism Statement

We hereby declare that the work in this graduation project at Yarmouk University is our own except for quotations and summaries which have been duly acknowledged. This work has not been accepted for any degree and is not concurrently submitted for award of other degrees. It is the sole property of Yarmouk University and it is protected under the intellectual property right laws and conventions.

We hereby declare that this report is our own work except from properly referenced quotations and contains no plagiarism.

We have read and understood the school's rules on assessment offences, which are available at Yarmouk University Handbook.

Students :

Sara sleit	Ansam Alawneh
	
11/10/2026	11/10/2026

1

Table of Contents

Students' Property Right Declaration and Anti-Plagiarism Statement……2

List of Tables……..……..……..……5

List of Figures……..……..……..……..…5

Abstract……..……6

Chapter One: Introduction…….…….....7

1.1 Problem statement and motivation……..…….....7

1.2 Background……..……8

1.3 Aims and Objectives……..……9

1.4 How has the problem been solved till now? ……..……10

1.5 What is our main solution idea? ……...…12

Chapter Two: Background ……..……50

2.1 An Overview of Previously Used Technologies for solving to Problem……..…32

2.2 context of the design……..……...……..33

2.3 Target market and there needs ……...…..19

2.3.1 target market.....19

2.3.2 target users and there need.....19

2.4 Potential ethical and environment issues……19

2.5 Summarize the different approaches ……..……..…22

Chapter Three: Design ……24

3.1 Design Overview……..…24
3.1.1 Description of the design……..…….....24
3.1.2 Planning to address the problem…….……25
3.1.3 Description of the interaction between the components Two Scenarios……...…26
3.1.3.1 High-Level Architecture Figure & System Layers……26
3.1.3.2 End User scenarios ……28
3.2 Design details ……..……..…….……..……30
3.2.1 Design Specifications ……... ……30
3.2.2 Design process and Detailed Techniques ……..……...……..31
3.2.2.1 Layer 1 & 2: Physical Data Acquisition & Edge Processing Subsystem ……31
3.2.2.2 Layer 3: Network ……34
3.2.2.3 Layer 4: Cloud Infrastructure, Analytics & Web Visualization Subsystem ……37
3.2.3 Legal Aspects…….…….....39
3.2.4 Design Constraints……..……...….40
3.2.5 Design Standards…….....41
3.2.6 Design Alternatives……..…….....42
3.2.7 Safety Consideration……..…….....43
3.2.8 Design considerations table……..…….....45

Chapter Four : Conclusion and Future Work…….46

4.1 Summarize the main contributions of the work 46
4.2 Further future work someone should do to make the system better……47
4.3 Lessons learned.....48

Reference…….49

List of Tables

Table 1 : 3.2.6 Design Alternatives……42

Table 2 : 3.2.8 Design Elements of the Suggested System……..……45

List of figures

Figure 1 : 2.1 Increase in the number of connected IoT (2012-2020)[9]……13

Figure 2 : 2.1 Different types of RFID tags [1]……15

Figure 3 : 2.1 MFRC522 RFID Reader Module with RFID Tag [1]……15

Figure 4 : 2.1 Two different types of force sensing resistor technologies [2]……16

Figure 5 : 2.1 Structure of a Capacitive Pressure Sensor[3]……17

Figure 6 : 3.2.2.1 HLK-LD2410B mmWave Radar [4]……31

Figure 7: 3.2.2.1 LD2410 Human Presence Radar Sensor Configuration Tool [5]……32

Figure 8 :3.2.2.1 Ceiling-Mounted Installation of the HLK-LD2410B Radar Sensor [6]…33

Figure 9: 3.2.2.1Wall-Mounted Installation of the HLK-LD2410B Radar Sensor [6]……33

Figure 10 : 3.2.1.1 INMP441 Microphone Sensor Module [7]……34

Figure 11 : 3.2.2.1 ME280 Temperature Sensor Module[8]……....35

Figure 12 : 3.2.7 Sealed Protective[10]……44

Figure 13 : 3.1.3 architecture of the proposed IoT-based Smart Library Monitoring System …27

Figure 14 : 3.2.1.1 ESP Block Diagram ……36

Figure 15: 3.2.2.1 Under-table mmWave radar setup for seat occupancy detection…… 34

Figure 16: 3.2.2.1 Sensor module with RF-absorbing shielding baffle to prevent side-lobe cross-interference……34

Abstract

This project develops an IoT-based Smart Library Monitoring System that uses ESP32 nodes and sensors to track real-time table occupancy and noise levels across library sections. Data is transmitted via Wi-Fi and MQTT to a central database, enabling a web dashboard with live statistics, heatmaps, and weekly/monthly analytics. The system applies predictive analytics to forecast peak usage periods, particularly during exam seasons, and includes an AI module for occupancy prediction, study area recommendations, and a chatbot. Automatic PDF reports support administrative decision-making. A comparative hardware evaluation ensures optimal sensor and microcontroller selection. The proposed system can serve as a scalable smart-building solution that can be integrated into broader Smart City platforms for intelligent facility management and efficient utilization of public spaces.

Keywords: Smart Library Monitoring System, Internet of Things (IoT), ESP32 Controller, Data Analysis and Prediction, Web Control Panel

Chapter One : Introduction

1.1 Problem statement and motivation

When visiting the university library or other study areas, it is sometimes noted that there is inconvenience or a lack of services accessible to provide a proper learning environment for students. There are challenges in meeting the needs of students when selecting a suitable place to study, such as searching between floors to find an empty seat, a quiet place, or a suitable place for group projects, in addition to taking into account the appropriate temperatures and the ease of searching for scribes to borrow, or taking into account people with special needs and providing designated and suitable places that are easily accessible to them.

These problems all stem from the reality that students face in university libraries. Academic libraries have made significant efforts to provide solutions to these problems and to create a suitable study environment for many years 4, since the lack of a suitable study environment significantly reduces student productivity, keeping in mind that the ideal environment varies from student to student based on their individual needs.

In light of this, attempts have been made to deal with these problems. At the Central Academic Library (Eindhoven University of Technology), 252 students participated in a study titled "What Matters for Students' Use of Physical Library Space?" The purpose of the study was to analyze student needs and the factors influencing their selection of study areas in academic libraries. The researchers found that space, noise level, crowding, comfortable furniture, and cleanliness were the primary factors that affected students' choice of study areas.

This study is related to the proposed smart library system because it shows that the availability of seats is not the only factor that influences students' choice of study spaces. As a result, the proposed system can build on this idea by implementing artificial intelligence and Internet of Things technologies to monitor seat occupancy and noise levels, display this information in real time, and offer students recommendations for ideal places and other options. This will help to develop the learning environment, maximize its use, and achieve the goal of serving both students and administration.

1.2 Background

Physical libraries in academic institutions are essential components that cannot be replaced by virtual solutions because they give students with access to academic material as well as a learning environment that encourages them to study, interact, and get through projects. To ensure that these places are used effectively, it is vital to research how students utilize them and determine their requirements.

Technological Transformation and the Emergence of Smart Libraries

With incredible developments in artificial intelligence and the Internet of Things, hundreds of devices are now networked, sharing data that can be analyzed and used to make decisions and make life easier. This has resulted in the replacement of systems that required manual labor. Libraries are among the most visible examples of this transformation, with easier and more simple access to information and services for students and administrative staff than ever before. This type of library, supported by modern technology, helps enhance the library's ability to meet users' needs.

The Core Challenge

Based on the above, the fundamental task is to change and develop libraries from traditional to smart libraries, replacing on-site monitoring and manual data collecting with an integrated system that offers real-time information. This is the perspective behind the "Smart Library Monitoring and Predictive Analytics System" project, which provides a comprehensive solution that integrates artificial intelligence and Internet of Things technologies , thereby ensuring more efficient use of study spaces.[3]

1.3 Aims and Objectives

The **Smart Library Monitoring and Predictive Analytics System** is an IoT-based intelligent engineering solution designed to optimize library space utilization and enhance students' study experience.

- Real-time table occupancy detection

- Noise level monitoring per library section

- Data storage and structured logging

- Accuracy testing and validation

- Predictive analytics for peak-hour forecasting

- Smart Reservation System

- Temperature Monitoring

- Visual web-based monitoring dashboard

- Accommodating the needs of people with disabilities

1.4 How has the problem been solved till now?

The problem of seat occupancy and space management in academic libraries has been addressed in several ways, from conventional systems to ones that employ AI and Internet of Things (IoT) technologies.

Manual and survey-based monitoring methods:

Some libraries have embraced traditional techniques to track the usage patterns of academic spaces by conducting periodic field observations inside study areas and manually recording student data. The main objective was to study students' behavior patterns and improve the study environment based on the findings.

Disadvantages and Limitations:

Although this data is useful for understanding general trends, this methodology has its disadvantages, including:

1. Time and Effort: Manual data collection and field trips require a lot of human effort.

2. Prone to human error: There is a significant chance that mistakes may be made when counting and recording student data, or that some information will be ignored or recorded twice.

3. Prone to a Lack of Real-Time Updates: Because these approaches rely only on reviewing prior data, they fail to provide students seeking suitable locations with real-time, up-to-date information.

4. Scaling Limitations: It is difficult to implement this method on a permanent basis due to the wide areas within libraries that are difficult to account for during peak hours.

[3][5]

Access Gate Logging and Perimeter Systems:

These systems rely on the installation of sensors, such as IR sensors, or card readers, such as RFID readers, at designated entry gates to count the total number of people inside the library or building.

Disadvantages of this method , including:

1. Do not give students or administrators any information about occupancy, noise levels, or temperature.

2. The number of library visitors does not equate to the number of occupied seats. Some students or academic staff may visit the library to borrow books or inquire about related information. [6]

Computer vision and camera systems:

The most famous methods that have solved the challenge of precisely identifying accessible seats yet conflict with the privacy of students or people within the buildings
Some buildings have placed security cameras within the halls, and the footage is analyzed to identify accessible seats and tables as well as track student movement in real time.

Among the disadvantages of this technique:

1.Invasion of Student Privacy
2. Cameras are expensive to use and process.
3. Optical blockage and environmental factors might impact information identification accuracy.

1.5 What is your main solution idea ?

The main concept behind the proposed solution is to create an intelligent system for monitoring and managing the library, so that the system collects data on the library's current status, processes it, and converts it into useful information—while respecting visitors' privacy and avoiding methods that might make them feel uncomfortable. This data is presented on an easy-to-use website that gives a clear picture of the state of the library.This system helps visitors make educated choices when choosing a study space that meets their needs and study style, and it also helps administrative staff understand typical library space usage and use the available data to improve facility management and provide a suitable environment for visitors.

Chapter Two : Background

2.1 Provide a background of the problem, in easy-to-understand terms

IOT :

With the rapid developments we are witnessing in the 21st century in information and communication technology and their impact on various fields, institutions, and sectors, there is a clear need for advanced technologies such as the Internet of Things to improve the efficiency of services and operations.

Pujar and Satyanarayana (2015) Pujar and Satyanarayana noted that the Internet has evolved from the “Internet of Communications” to the “Internet of Things,” facilitating the connection of objects and the transmission of data—with or without direct human-computer interaction. While the Internet of Communications contributed to improving communication and services, the Internet of Things has enabled the connection of objects through the use of sensors and network capabilities.[9]

The Internet of Things is a term composed of two words: “Internet” and “things.” The Internet is a global, integrated system that connects computer networks to one another and helps users access information quickly and easily and retrieve it from anywhere. “Things” refer to any physical object or device in the real world.

The Internet of Things relies on a cloud platform that connects things to one another in the real world.

The term was first used in 1999 by British pioneer Kevin Ashton, who defined it as a system that connects the Internet to the real world through ubiquitous sensing devices. [10]

The Internet of Things will also revolutionize the field of libraries and information, so librarians must be familiar with IoT applications and services in libraries.[10]

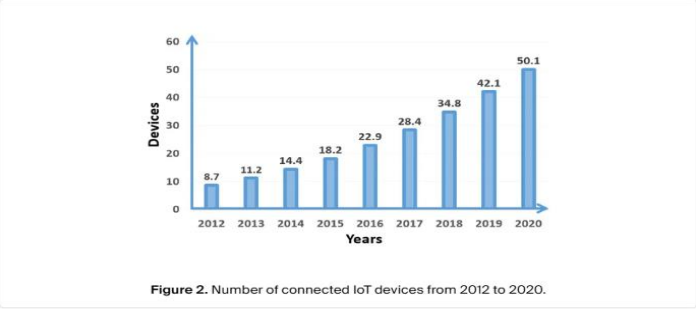


Figure 1 : 2.1 Increase in the number of connected IoT (2012-2020) [1]

Ai :

In recent years, the concept of artificial intelligence has become widespread and has emerged as an essential tool in various fields, including libraries.

Artificial intelligence refers to the ability of computer systems or machines to simulate human-like thinking and learning processes. It relies on artificial neural networks that mimic the processing capabilities of the human brain, endowing them with a high level of cognition and analysis [11].

The integration of artificial intelligence into Internet of Things (IoT) systems—often referred to as Artificial Intelligence of Things (AIoT)—has garnered significant attention due to its effectiveness in improving performance and the efficiency of intelligent decision-making. A key role of artificial intelligence is its ability to process large volumes of data [12] and analyzing it in real time, which helps the Internet of Things efficiently handle the continuous flow of data and increases the accuracy and reliability of the Internet of Things by identifying complex patterns that are difficult for humans to understand. This integration paves the way for new innovations and enhances the interaction between humans and their physical environment [13]

RFID:

Radio-frequency identification (RFID) technology, which allows for the wireless identification of products and simultaneous identification of multiple tags at once without manual developed in recent years from a specialised, narrowly targeted innovation to a key effort, has component for improving the handling of goods and produced materials.

Radio-frequency identification (RFID) technology, which allows for the wireless identification of products and simultaneous identification of multiple tags at once without manual effort, has developed in recent years from a specialised, narrowly targeted innovation to a key component for improving the handling of goods and produced materials.

Without requiring direct physical contact, objects can be identified using Radio Frequency Identification (RFID), a wireless technology that automatically gathers data via radio waves.

Its benefits include less dependence on handiwork, the removal of human mistake, and the availability of readily available data; additionally, RFID signals can contain things like plastic and wood, but not metals.

Although there are a variety of types of RFID, they can be broadly classified into two categories:

Initially:

Active Tags: These need a power source, which can be from built-in batteries or a connection to a powered system. Depending on the stored energy and the frequency of read operations, their operating lifespan is limited.

Secondly:

Passive tags have an endless lifespan and don't require batteries or maintenance. The three primary parts of these tiny devices are an antenna, a semiconductor chip, and a protective shell. The reader is the source of their strength.

There are two main types of energy transfer from the reader to the tags, depending on the Near Field and Far Field features of electromagnetic waves:

electromagnetic wave capture and magnetic induction. [17]

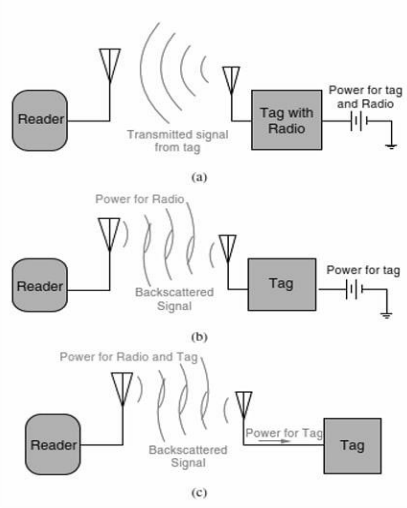


Figure 4. Different types of RFID tags: (a) active, (b) semi-passive, (c) passive [108]

Figure 2 : 2.1 Different types of RFID tags [1]

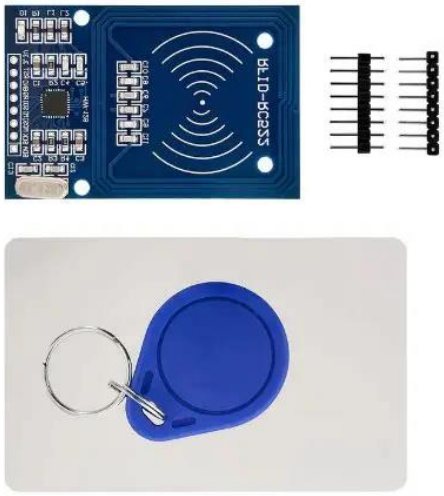


Figure 3 : 2.1 MFRC522 RFID Reader Module with RFID Tag [1]

FSR SENSORE:

A resistor whose resistance (measured in ohms) changes according to a pressure condition is known as a force-sensitive resistor (FSR).It is defined by being simple to use, cheap, small, and easy to set up and connect to microcontrollers like the ESP32. An FSR sensor's resistance changes with pressure; as pressure increases, resistance decreases, and the other way around.On the other hand, in the absence of pressure, the resistance is very high and near infinity, which appears as an Additionally, they can differ by up to 10% amongst open circuit. They are rarely accurate, though. as a result, you should anticipate particular response ranges while employing FSR sensors. They sensor are capable of measuring weight, however they are not the ideal option for precise weight determination.[18]

A semiconductive substance or conductive ink is positioned between two thin layers to form an FSR sensor.

FSR sensor technology fall into two categories:

First, there are shunt-mode FSR sensors, which are made of two membranes with a narrow air gap between them and a thick polymer layer.

Configuration: One membrane is coated with a unique resistive ink with a unique texture, while the other membrane features a set of interleaved traces that are electrically isolated from one another.

Thru-Mode FSR sensors come in second. The two outside layers of these circuits are made of polyester film and are referred to as flexible printed circuits.

Configuration: A pressure-sensitive layer and a conductive polymer layer are positioned above and below silver circuits with traces, and the two substrate layers are joined by an adhesive layer .[19]

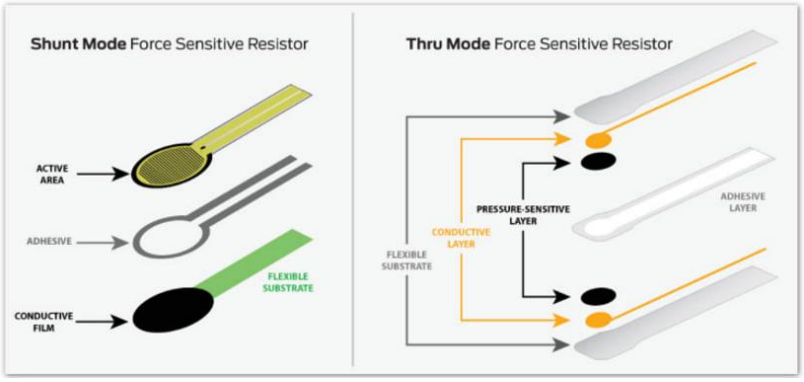


Figure 4 : 2.1 Two different types of force sensing resistor technologies [2]

Capacitive pressure sensors

Capacitive pressure sensors:

converts pressure into corresponding electronic signals by measuring changes in electrical capacitance caused by the pressure.[20],[21]

Capacitive pressure sensors are characterized by their light weight, flexibility, high accuracy, and fast response, which makes them suitable for use in wearable devices. They are capable of sensing various types of materials, are low-cost, have low energy consumption, a distance . They are also highly can operate without physical contact, and can sense from sensitive, capable of measuring relatively weak pressure signals.[20] , [21]

The performance of flexible pressure sensors is closely related to their structural characteristics and components.

Depending on the sensing mechanism, flexible pressure sensors are classified into three main categories: capacitive, piezoresistive, and piezoelectric. When comparing these three types, the capacitive pressure sensor stands out for its good stability, simple structure, and low power consumption.[22]

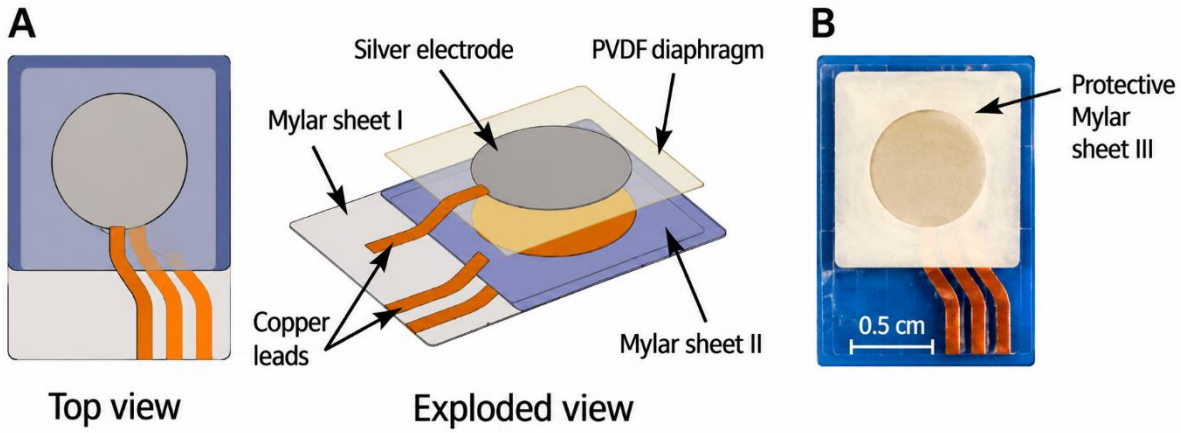


Figure 5 : 2.1 Structure of a Capacitive Pressure Sensor[3]

2.2 Provide some context about the problem, and why it is important and where it is used

Considering the fact that humans spend 90 percent of their time indoors, it is essential to create a healthy and comfortable indoor environment for them. Humans form the basic foundation of "Human-Cyber-Physical Systems" and occupancy data is essential for the "Post Occupancy Evaluation" of buildings. The purpose of the Post Occupancy Evaluation of buildings is to evaluate the usage of the space and measuring the indicators such as mobility and waiting time.

The paper "Seat Selection Behavior for Individual Study in a Digital Reading Room" discussed an important matter concerning the arrangement of space that could optimize limited space and increase users' satisfaction. In libraries, there have been efforts to prolong the stay of students in such spaces by recognizing spatial qualities that are suitable for study or understanding the preferences of visitors in communal corridors. However, since people have different habits and learning styles, more studies on what kinds of spatial qualities should be considered are necessary. One of these qualities is "personal space," which is explained by Katz (1937) as an invisible and mobile boundary around an individual that cannot be violated from a social point of view.

This is done so that people can stay away from stress-causing elements; therefore, the readers in the library look for their own space in order to stay away from any sort of disturbance when reading. In addition, the preference of the students about being closer to windows and lights is very particular; that is, those who like to read in single rooms want to sit with walls on both sides, and those who prefer to sit at open tables want a wall behind them. [15]

Another research paper, named "Occupancy Patterns in University Libraries in the Post-Pandemic Era," has been done on the seat reservation system in the university library in one of China's leading universities. The researchers have considered the library of a university providing education to a student population of around 60,000 from different disciplines. The study reveals three peak hours of occupancy, namely 11:00 am, 5:00 pm, and 8:00 pm, during which the libraries experience a higher rate of occupancy [14].

During the exam season, the occupancy rate increases as students need a quiet place to study; however, this is the time when there is an activity known as "phantom booking." In phantom booking, students keep their belongings on the seat and leave the library premises [16].

2.3 Target Market and User Needs

2.3.1 target market

The suggested system's primary target market includes universities and colleges with libraries and study areas but encounter issues in delivering an appropriate and user-friendly environment for administrators and students.

2.3.2 target users and there needs

The target system addresses the operational requirements of two main user categories

student :

Identifying available spaces and making it easy to find a suitable location based on each student's needs, supplying students with information on temperature, noise,and construction work improving the student's learning experience and increasing their productivity.

The system was also designed with users with special needs in mind to ensure their easy access to learning spaces and to provide simple and straightforward ways to access system data

library staff:

Monitor library operations to study how library spaces are utilized, identify peak times at the library, and make decisions based on the information provided in the reports generated, thereby facilitating the identification and resolution of problems and the study of students' library usage patterns.

2.4 Potential ethical and/or environmental issues.

The use of Internet of Things (IoT) technologies is essential in buildings because of their impact on improving performance and utilizing the data collected. However, certain challenges must be addressed regarding the social, ethical, and environmental issues arising from the use of these technologies in buildings:

1. Differential Privacy (DP)

It is a mathematical algorithm used to protect data. It works by adding noise to the real data collected from an Internet of Things (IoT) system, with the goal of preventing access to the real data .

However, this algorithm causes problems because the system is highly sensitive to the presence of people and uses real-time data; therefore, adding false data can affect the system's efficiency, the accuracy of the information, and energy consumption, as the system may mistakenly detect the presence of people and turn on the lights or adjust settings based on incorrect information

2. Behavioral Suppression and Psychological Impact

When students and staff feel they are constantly being monitored by motion sensors or cameras, their natural behavior changes and they do not feel at ease.

3. Risks and Physical Security of Data Storage [7]

This concept relates to the physical security of the building and its contents, and to maintaining that security from within

For example, if information regarding the building's occupancy status (such as the absence of people or empty rooms and sections) were to fall into the hands of dishonest individuals, they could exploit this information to steal physical resources such as equipment, books, or student property.

4. Using Internet of Things (IoT) Devices as Entry Points

IoT devices are connected to Wi-Fi networks, and because the data generated by these sensors and controllers cannot be adequately protected—since it is not possible to install security protocols on them due to their limited capabilities—wireless attacks on them provide hackers with a gateway to access more sensitive information related to the university and other large servers

5. How and Where to Store Data

1. Wired Connection Without a Cloud

This method involves connecting sensors via cables directly to a local server within the library.

One of its advantages is that it is very secure, but it requires expenses for digging and cabling.

2. Using a Cloud Server

This approach involves sending data to an external server such as AWS or Azure. One disadvantage is that you must ensure the reliability of the external company that will manage this data, which requires implementing highly secure authentication agreements with these companies.

6. Sustainability of Devices Used in the IoT System

The use of dozens of sensors, microcontrollers, and batteries continuously consumes electricity and generates electronic waste that harms the environment We must learn how to manage and recycle this waste and find solutions to reduce energy consumption and promote energy conservation. .

2.5 Summarize the different approaches

Many studies have focused on academic libraries and improving their performance by proposing and implementing solutions, including the following:

1. Traditional Methods

Torres and Paul used traditional methods within libraries to collect information on how the library was used over a specific period. A team recorded this information to help improve library services and study students' usage patterns in order to make the library a more suitable environment.

The data collection process was entirely manual; they recorded students' seating locations and seat numbers, as well as the time the data was collected. Among the tools they used were:

Excel, MS PowerPoint, and Qualtrics.

However, this method presented some challenges, including: counting errors, a lack of real-time information, and dependence on human observation.[1]

2. Pressure Sensors (FSR) + RFID + WEB APPLICATION

DANIEL proposed a design for the library that allows students and library staff to access information about library occupancy while also identifying the person seated there

This information can be accessed both inside and outside the library via the Internet using smart devices.

The second way to access the information was through a screen displaying seat availability at the library entrance.

This proposal helped reduce the time spent searching for an available seat inside the library, but there was no way to control who could view this data. Furthermore, it provided information only on seat availability and the user's name, without ensuring privacy.

Thus, privacy and visibility are among the most prominent issues.

3. RFID + FSR + IoT

The system was later developed under the name “IoT,” with the addition of a reservation feature that allows students or researchers to see which seats are available and reserve them before arriving at the library, in addition to displaying real-time information on a website accessible from anywhere.

Further challenges arose, such as ensuring that the student who made the reservation would actually show up, and if they did not, how to update the seat’s status to “unreserved.” Or, if the seat was occupied by an object rather than a student, how to handle the data generated by the FSR—specifically, how to process such instances.

4.Capacitance Sensors

In order to collect information, improve library services, and analyse student usage patterns with the aim of resolving the problem of seat exclusivity and improving the learning environment, researchers employed a system based on capacitive sensors to monitor seat occupancy in libraries over an 8-week period, covering 36 seats. Periodic voltage readings from the sensors were used in the data collecting procedure, and seat status was tracked and directly updated online. A database (MySQL), microcontrollers (Raspberry Pi), capacitive sensor chips (such the MPR121), and a web application for showing real-time maps were some of the tools and technology they employed. However, the technique experienced a number of difficulties, such as sensor readings that changed as a result of shifting surroundings, signal level reduction brought on by extended cable lengths, and device problems.[33]

5. Computer Vision / AI

In order to maximise the use of library seating resources and reduce the stress for administrative staff, some researchers have employed computer vision techniques and deep learning algorithms to monitor seat occupancy in libraries during exam periods via surveillance cameras installed in the reading rooms. A smart video processing unit, a server-based backend unit data storage, and a display unit through a WeChat mini-program made up the system. However, the system's disadvantages include its expensive cost and attacks on others' privacy . [32]

Chapter Three : Design

3.1 Design Overview:

In this project, we are looking for a solution to a number of concerns that have a negative impact on both library patrons and administrative personnel, including congestion, challenges in organising study spaces, and excessive noise levels. Standard methods have not been able to handle these problems.

In order to provide a suitable, appropriate, and comfortable study environment, the project was designed to create a smart engineering system for library management that would address these issues at the lowest possible cost, lessen the workload for administrative staff, anticipate peak times—especially during periods of high congestion, like exam periods when library occupancy rates are very high—and suggest a seating arrangement for each visitor based on their individual or group study preferences.

3.1.1 Describe the design of your project to achieve the solution：

A smart monitoring system of the library comprises several essential components which are supposed to accomplish complementary tasks as follows: Millimeter-wave radar sensor is supposed to monitor seat occupancy due to emission of high-frequency waves which will be used to detect people as well as their minor movements without any physical interaction and influence on privacy, whereas temperature sensor is aimed at measuring the temperature in the area in order to get the data on the library's environment, and the sound monitoring sensor measures the sound level in the area. Central processing unit acquires the data from the sensors and carries out preliminary processing of them prior to transfer via the wireless network to the central server system, thus receiving, processing, organizing and managing the data in order to connect the database with the user interface. Database helps to store information effectively, facilitating its quick access and modification. At the same time, the web-based user interface demonstrates this information, thus helping the library visitors and administration to track the current state of the library quickly.

3.1.2 How you plan to address the problem statement.

Based on the challenges and problems faced by both library visitors and the library's administrative staff, this system offers technical solutions to address these issues with high efficiency**:**

1. Eliminating Random Searches and Saving Time:

The system eliminates the randomness of finding seating by providing a live, interactive map that reflects real-time occupancy status within the library, thereby reducing the time and effort students spend finding a suitable place for their study style.

2.Environmental Control:

The system relies on a network of sensors to measure temperature and noise levels in the areas surrounding the sensors throughout the library and alerts visitors to help minimize noise disturbances.

Data is displayed on a control panel and the website, assisting visitors in selecting a space that meets their environmental preferences.

3. To minimize false readings and protect privacy:

To avoid the use of cameras that infringe on visitors' privacy, a secure alternative has been adopted in the form of capacitive pressure sensors to monitor seating, distinguish between objects and visitors, and prevent "phantom" reservations

4. Improving Management and Decision-Making:

The system collects and stores data related to seat occupancy and environmental conditions for analysis, enabling the study of library usage patterns and peak times.

Artificial intelligence helps generate predictive analytics for periods of high traffic and supports management in making accurate, data-driven decisions, while the system also supports the generation of periodic reports.

5. Ensuring System Integration and Ease of Access:

The system is designed to operate seamlessly by connecting the main processing unit to the central database via a wireless network, ensuring real-time data synchronization between the library's sensors, the user app, and the administrative control panel. This ensures seamless integration, rapid system response, and minimizes errors and data transmission time.

3.1.3 Provide a detailed figure for solution and Two Scenarios

3.1.3.1 High-Level Architecture Figure & System Layers

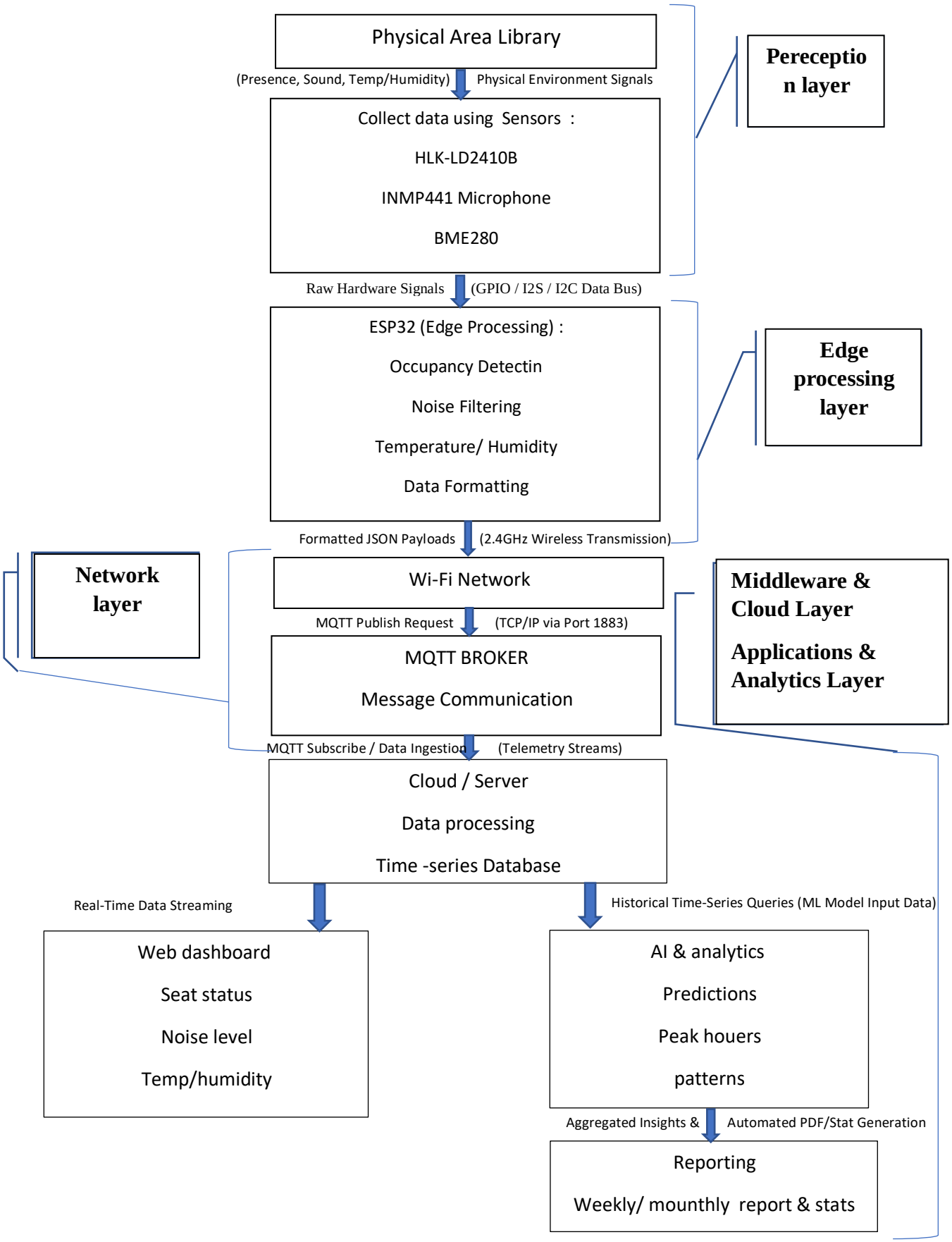


Figure 13 : 3.1.3 architecture of the proposed IoT-based Smart Library Monitoring System

Architecture of the suggested system is designed as a layered IoT architecture for monitoring and analyzing the environment of the library. This architecture depicts various layers of the system along with interactions between them beginning with the process of data gathering using the sensors and edge computing with ESP32 module, then data transferring via WiFi and MQTT, followed by storing, processing, analyzing, and visualizing of data. Every single layer has its own job and interacts with the neighboring layers in order to provide a seamless data flow.

Architecture includes five main layers, each being dedicated to a particular stage of system processing. This section describes the purpose of each layer, the elements it contains, and its **main** functions.

Pereception layer :

It is the first layer of the typical IoT model and is referred to as the sensor layer. The function of this layer is gathering information from the environment in which the IoT model is implemented. Information is gathered using sensors that make up IoT models. One of the notable characteristics of sensors is addressability that allows sensors to have addresses via which we can communicate with them. The second one is sensing and activation. It has already been stated that sensors can be identified as devices capable of gathering information about the surrounding environment [23].

The sensors used in this system are:

1. **HLK-LD2410B 24GHz mmWave Human Presence Radar Sensor Module**

This particular sensor detects if a student is seated at their desk in the library.

2. INMP441 Omnidirectional MEMS Microphone Sensor

This is a sensor that measures the sound level in each library section .

3. BME280 Environmental Sensor

This sensor measures: Temperature ,Humidity, Atmospheric pressure

Edge processing layer :

The edge processing layer is responsible for performing initial processing of the data collected from the sensors; this process involves simple data processing before the data is sent to the server. In this proposed system, the ESP32 acts as an edge processing unit, receiving data from the sensors and performing simple processing on it. The goal of this processing is to reduce the amount of raw data streamed to the server and to minimize the load on the network and the server. After this stage, the data is transmitted to the network layer.

Figure 13 : 3.1.3 architecture of the proposed IoT-based Smart Library Monitoring System

Network layer :

This layer transmits data from the ESP32 modules to the server via a Wi-Fi network, while the MQTT protocol is used to exchange messages between the ESP32 modules and the MQTT broker, which acts as an intermediary to receive the data and forward it to the server.

Middleware & Cloud Layer

The objective of this layer is to collect, store, and process data that comes from ESP32 modules. The middleware & cloud layer comprises several elements such as MQTT Broker, Time-Series Database, and Data Processing Engine. Once the data is processed and stored, it becomes available to be presented and used by users through the Applications, Analytics & Artificial Intelligence Layer.

Applications & Analytics Layer

The role of this layer is to function as an interface for interaction with the system as well as providing users with advanced analysis of data and predictions based on the collected data.

The Applications & Analytics Layer includes the following elements:

- Interactive Web Dashboard

Displays seat status, occupancy, noise levels and environmental data

- AI & Predictive Analytics Module

Analyzes data and provides predictions on usage patterns and peak hours

- Reporting System

Generates weekly and monthly reports and statistics

3.1.3.2 End User scenarios

This clearly outlines two scenarios for how end users (students and library staff) use this system.

Scenario 1: Student Seeking a Suitable Study Place

A student is trying to find a suitable place to study within the library. However, before getting into the library, the student accesses the Smart Library Monitoring System through the web dashboard.

1. The student opens up the web dashboard where he/she can check the status of the library sections.

2. The dashboard provides information about availability of seats, noise level, temperature, and humidity of the respective sections.

3. The student makes comparisons among the available sections and decides on a suitable study area from the given information.

4. All the sensor data are continuously gathered, processed, and then sent to the server via the ESP32 modules using Wi-Fi and MQTT.

5. Dashboard updates itself with the latest information so that the student makes the selection according to the present condition of the library.

6. Predictive analytics also provides information regarding future occupancy or peak times for the library sections.

Scenario 2: Library Management Monitoring and Decision-Making

The administrator of the library wants to monitor the utilization of various library sections and identify periods when there is high occupation or noisy environment.

1. The administrator logs into the web dashboard to monitor the current state of the library.

2. The system presents the data regarding seat occupancy, noise level, temperature, and humidity of various library sections.

3. The administrator monitors the data that has been gathered using the web dashboard and identifies sections where there is high occupation and/or noise.

4. The system gathers the historical data in the time series database.

5. The system applies artificial intelligence and predictive analytics to analyze the gathered data and make predictions about high-usage periods.

6. The reporting system provides statistics on weekly and monthly basis that assist the administrator in evaluating the usage and environment of the library.

7. Based on the data and predictions the administrator makes decisions about library management and utilization.

3.2 Design Detail

This section explains the detailed design of the proposed system. It includes the design specifications, implementation process, constraints, standards, alternatives, legal aspects, safety considerations, and other important design factors

3.2.1 Design Specifications

To ensure the system's efficiency and practical integration within the library environment, a set of criteria was taken into account, including aesthetic, environmental, and economic aspects, as outlined below

1. Dimensions and Aesthetics:

The sensor units feature a compact, integrated design that allows for discreet installation out of users' sight, thereby protecting the equipment from accidental tampering and preserving the library's aesthetic appeal.

2. Environmental Conditions and Work Environment Suitability:

The system is designed to operate efficiently and takes into account internal environmental changes in a library, such as temperature, ensuring uninterrupted operation while providing a comfortable environment for users.

3. Cost and Maintenance:

The system components were selected because they are relatively low-cost and energy-efficient compared to traditional solutions, such as camera systems. They are also easy to install, maintain, and troubleshoot with minimal effort.

4. Accessibility and Comfort of the Space:

The system's user-friendly, simple web interface is made according to the different requirements of library users. It has been suggested that the bottom level be set aside for those with special needs in order to make it easier for them to access seating spaces and utilise library resources while also ensure their comfort.

3.2.2 Design Process & Detailed Techniques

3.2.2.1 Layer 1 & 2: Physical Data Acquisition & Edge Processing Subsystem

A. Sensor 1: HLK-LD2410B mmWave Radar (Presence & Occupancy Logic)

The human presence sensor can detect the presence of a person, whether they are moving or stationary. This sensor operates on the principle of 24 GHz mm-wave radar and uses FMCW radar technology.

It was selected based on its many advantages, which addressed gaps in previous sensors. Its advantages include:

1. Detection of both moving and stationary people: In addition to its high sensitivity to moving objects, this sensor can also detect stationary people.

2. Ability to conceal it within a non-metallic housing:

This sensor's ability to penetrate non-metallic surfaces helps improve the final design aesthetics upon installation.

3. This sensor is distinguished by its ability to adapt to environmental conditions: It is not affected by lighting or temperature.

4. Adjusting the detection range and sensitivity for people: With this sensor, we can easily adjust the detection range and set the sensitivity for each zone to prevent interference between sensor ranges, as each sensor should detect only a specific zone.

5. It includes a delay time adjustment feature: Through the sensor settings, we can set a delay time for vacating a seat. For example, if a student needs to leave to borrow a book or for another reason, the seat's status will not change immediately upon their departure; an appropriate vacating time will be set.

[26]



Figure 6 : 3.2.2.1 HLK-LD2410B mmWave Radar [4]

Pins and Their Functions for This Sensor

1. OUT PIN: This pin is the target status output; that is, it is responsible for transmitting the seat's status directly and digitally.How this pin works: When a person is present, this output sends a high-voltage signal; when no one is present, it sends a low-voltage signal.

2. GND PIN: Completes the electrical circuit

3. VCC PIN: The power supply input for the sensor to enable it to operate;it is connected to a 5V power source.

The manufacturer has integrated Bluetooth with the sensor, allowing us to configure and calibrate the sensor via the HLKRadarTool mobile app using Bluetooth, rather than using the UART-RX and UART-TX ports. This makes connecting the sensor to the microcontroller easier and simpler.Therefore, we will limit ourselves to using only these ports: VCC, OUT, GND.

Through the software, we will configure the following: maximum range for motion detection, maximum range for detecting stationary people, and evacuation delay time. The following figure illustrates the software and its flexibility in changing the sensor's settings:

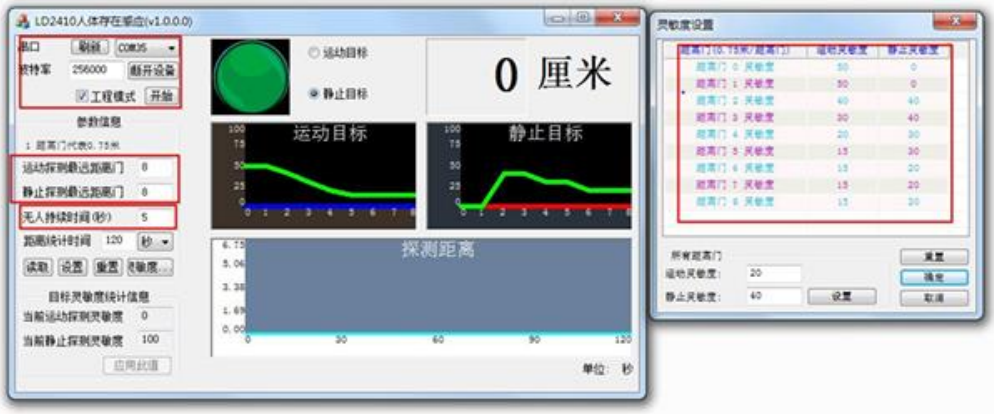


Figure 7: 3.2.2.1 LD2410 Human Presence Radar Sensor Configuration Tool [5]

Installation Procedure:

1.Company's suggested installation(HI-LINK) [26]:

The area to be monitored is covered in a conical pattern using two installation methods: on the ceiling of the room or on the side of the wall. However, this method is not practical for our system because it does not specifically detect which seats are occupied; it only detects whether or not there is a person in the specified area:

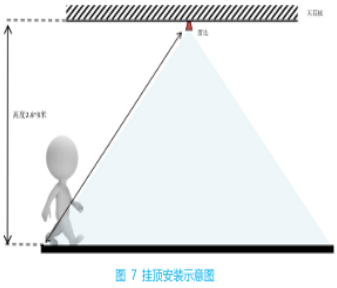


Figure 8 :3.2.2.1 Ceiling-Mounted Installation of the HLK-LD2410B Radar Sensor [6]

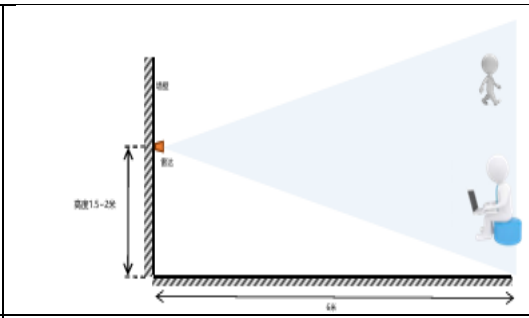


Figure 9: 3.2.2.1Wall-Mounted Installation of the HLK-LD2410B Radar Sensor [6]

Our system proposal: This proposal utilizes the same sensor but in a way that allows the status of each seat to be determined individually to obtain detailed information. The proposed installation process will be as follows:

The idea is based on the following principles:

1. Assigning a separate sensor to each seat: to obtain more accurate information about the status of each seat and display its status on the dashboard separately and independently;

2. Directing each sensor toward the seating area of the seat it monitors; designing a metal shield or enclosure that prevents the transmission and reception of signals from adjacent areas and focuses solely on the target area.



Figure 9: 3.2.2.1 Under-table mmWave radar setup for seat occupancy detection.[ai]

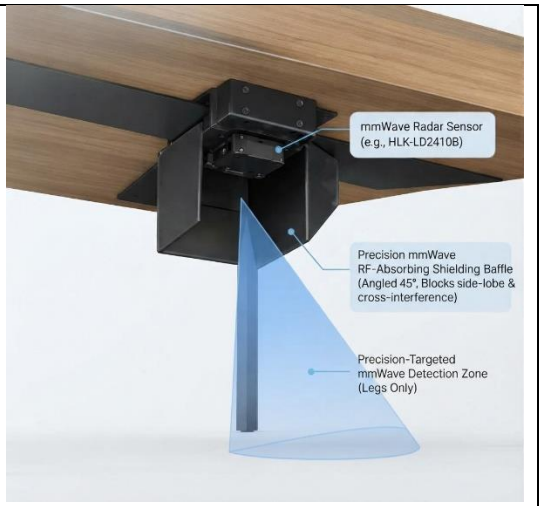


Figure 10: 3.2.2.1 Sensor module with RF-absorbing shielding baffle to prevent side-lobe cross-interference.[ai]

B. Sensor 2: INMP441 MEMS Microphone (Noise Processing & Privacy)

It's an electronic microphone whose function is converting sounds to digital data, which are then sent to the ESP32 for processing. In our project, it's meant to be used in measuring the noise levels within the library through using MEMS technology. It has the following features that made us select it:

1. Power efficiency: One of the most notable feature in IoT devices, since this particular microphone uses very little energy when working.

2. Omni-directional: Made in a way to record sounds coming from different directions and not only one for giving us precise noise levels data.

3. Interface of I²S and 24-bit resolution: The microphone doesn't convert sound to the audible audio directly to the ESP32, but first to digital data.[27]

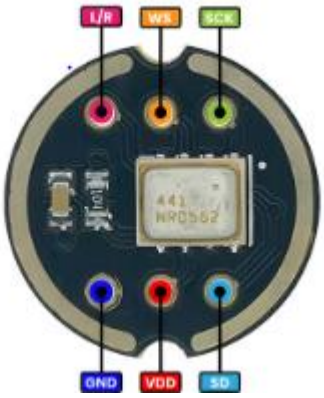


Figure 10 : 3.2.1.1 INMP441 Microphone Sensor Module [7]

As to the function of pins, here they go:

VDD – provides power for microphone

GND – the pin that finishes the circuit line

SCK – clock pulses, necessary for timing of the audio data transfer using the I²S protocol

WS – pin for selection of left/right channel in the I²S stream

SD – Serial Data pin – the key one that provides data transfer of the digitized audio samples from the environment

L/R pin – the pin for selection of the left/right audio channel in the I²S stream; connected to GND to select left one.[27]

C. Sensor 3: BME280 Environmental Sensor (Climate Monitoring)

It is a sensor that measures humidity, atmospheric pressure, and temperature. It uses the I2C protocol, and the pins are connected as follows:

SCK pin: Connected to the clock line of the microcontroller to control the data transfer timing.

SDI pin: Sends data to the microcontroller.

VIN pin : provides power for sensor

GND pin : the pin that finishes the circuit line [28]



Figure 11 : 3.2.2.1 ME280 Temperature Sensor Module[8]

Key Features:

1. Integrates three core functions into a single chip
2. Supports the I2C protocol, which simplifies installation for the user without causing conflicts
3. Includes ready-to-use ports that require no soldering
4. Low power consumption[28]

Principle of Connecting Sensors

(BME280 Environmental & INMP441 MEMS Microphone)

To measure temperature, humidity, and pressure, as well as noise levels, an appropriate number—ranging from 2 to 3—of these sensors will be installed in each room. We do not need a large number of them; rather, we need to distribute them appropriately within each room and calculate the average of the results

To optimize resource use, these sensors will be installed in our system on the same ESP32 module located under the table, along with the occupancy sensors. As mentioned, this will be implemented on only 2–3 tables in each section to calculate the average.

D. ESP32 Edge Gateway & Hardware Interfacing (I2S, I2C)

All data collected from the sensors will be transmitted to the ESP32 module, which performs preliminary data processing.

Examples of the processing it performs include the following: [25]

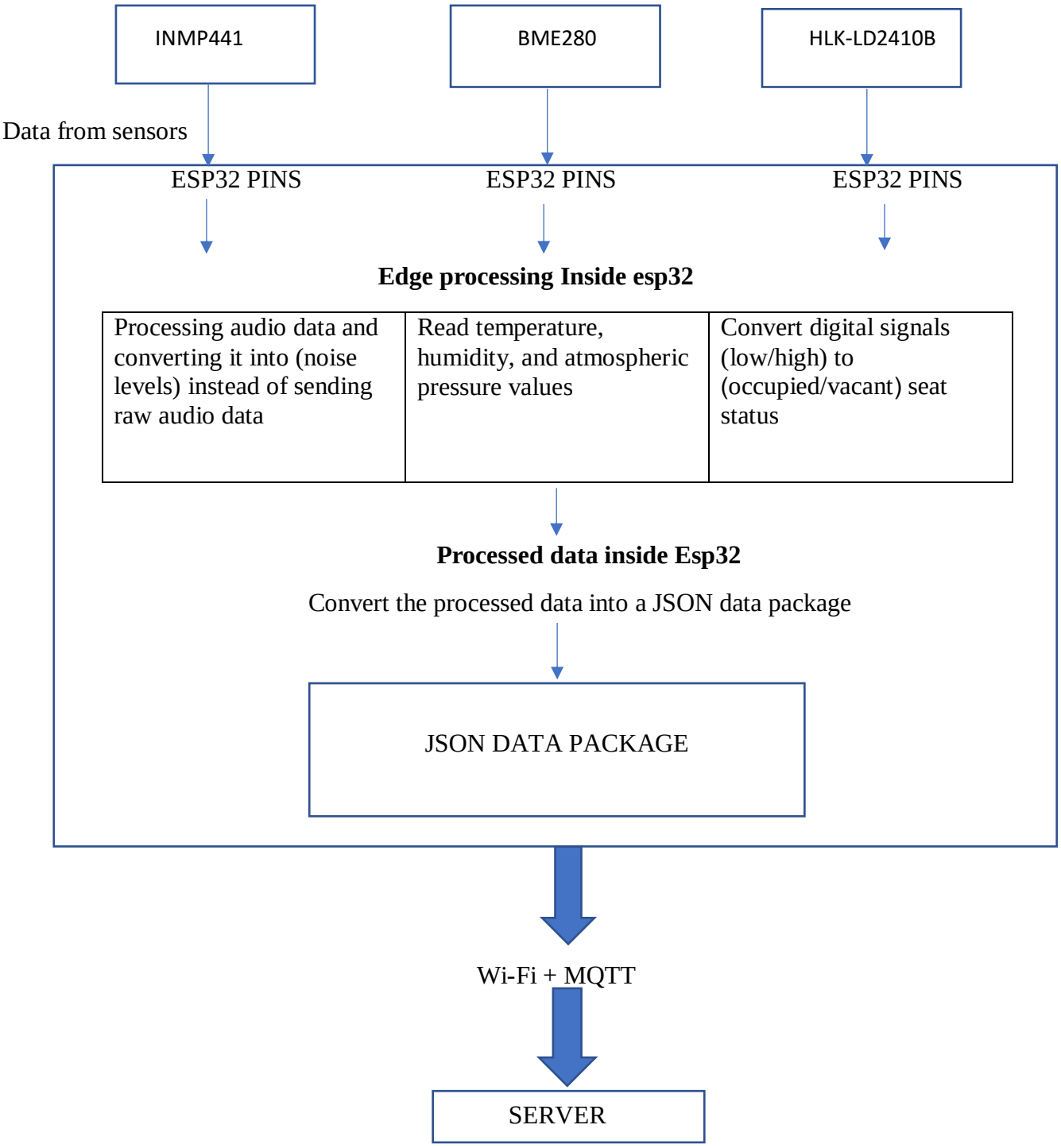


Figure 14 : ESP Block Diagram

3.2.2.2 Layer 3: Network

The ESP32 module uses its built-in Wi-Fi chip to connect to the Internet, and an encrypted library is used to establish a secure connection with the cloud server.

This is done by converting the data collected from the physical environment into JSON format; this data is then encrypted before being transmitted via the antenna, which allows us to protect the data from hacking.

The MQTT protocol used to transmit this data is a lightweight protocol employed in the Internet of Things (IoT) and its applications. Its purpose is to connect devices with limited capabilities—such as the ESP32 and sensors—to the server over networks.

It consists of three components:

Publisher: In this project, the publisher is the ESP32.

Broker: This is the server that receives messages and routes them.

Subscriber: In this project, it is the dashboard.

3.2.2.3 Layer 4: Cloud Infrastructure, Analytics & Web Visualization Subsystem

A. Cloud Server and Backend Logic:

Once data is sent from the sensors to the ESP32 and then routed through

MQTT Broker

located in the cloud, this data is captured by a backend application.

This application is programmed using multiple programming languages such as:

Node.js and Python.

The backend service serves as the system's primary processing motor.

B. Data Processing on the Server, Analytics and Predictions

When the backend receives the data it looks at the JSON and does real-time aggregation. This step needs heavy data processing to ready the data. It creates information for each section of the library by calculating the average sound sensor readings in each section the occupancy rates and the temperatures in each section. It also decides how this information will appear on the dashboard.

For analytics the backend keeps all sensor data in databases for later analysis. The backend script uses programming languages, special analytics libraries and predictive algorithms to examine the data. For example the system may notice that a particular library is always busy at 11 p.m. From that it makes a forecast that is shown to students and administration.. If the system sees that a seat stays in the same status it predicts a malfunction with that seat and sends an alert to administration so the issue can be checked and fixed.

As for statistics an automated script on the server scans the database and pulls data to make statistics about the library's daily and weekly occupancy rates, peak times average temperatures and energy use.

Finally the server exports the needed data, as a ready-to-use file. Sends it to administrators.

C. Interactive Web Dashboard

After all the preceding processes—from the sensors to the final data processing on the server—the web dashboard comes into play

to serve as the user interface and visual display layer for the smart library system.

It is responsible for:

1 Displaying the status of each seat in the library by changing the chair's color (red for occupied, green for available), based on data provided by the server;

2 environmental indicators and analytics;

3 graphs and digital dashboards displaying key metrics;

and displaying expected peak times for student traffic.

3.2.3 Legal Aspects

In order to protect user data and avoid data misuse, the benefactor of the "Smart Library Management System" follows with ethical, legal, and privacy standards. The following examples show how these standards are being followed:

1.Obedience to the General Data Protection Regulation (GDPR):

This law, which was established by the European Union to protect personal data, specifies the procedures that businesses must follow while gathering, storing, and processing data.[29]

These guidelines were followed in the creation of the Smart Library system, guaranteeing that no personally identifiable information (PII) about users is gathered or kept.

2 .Intellectual Property Rights and Software Licenses:

The system ensures compliance with the source code for microcontrollers and the control interface by relying on open-source tools and libraries that respect intellectual property laws.

3. Legal Responsibility for Data Security:

The system is designed to apply the highest security standards by routing all data transfers and the processing of sensitive readings through protected and secure channels to protect the system in the future from any security attacks or access without permission

4. Safety and Wireless Communication Standards:

The system uses wireless components that conform to standards set up by telecommunications regulatory organisations and function within legally allowed frequency ranges.

3.2.4 Design Constraints

We discovered specific limits and constraints during the system's design, which were taken into consideration. These limitations can be summarised up as follows:

1. Hardware and Technical Limitations:

The two primary technical issues that the project experienced were related to the microcontroller's capabilities and the process of choosing the human presence sensor.They were dealt with as follows:

1.1 Microcontroller Challenges:

These include the ESP32 microcontrollers' limited memory and processing speed, which requires the development of short code to ensure system stability and energy efficiency.

1.2 Difficulties in Choosing a Human Presence Sensor:

- The first sensor that was suggested was the Force-Sensitive Resistor (FSR) Sensor, however it lacked the necessary accuracy for human detection.
- Capacitive Pressure Sensor: Located unde rthe seat, this sensor needs to be linked to the main processing unit.Placing a specialised primary processing under each seat was the solution because the seats are not fixed and are Often relocated; however, putting this concept into effect is expensive and resource-wasting.
- Capacitive Proximity Sensor: This desk-mounted sensor has the disadvantage of wasting resources and not providing accuracy data for every student seating position.
- The infrared motion sensor (PIR), which is positioned under the desk, has the disadvantage of not being able to identify stationary objects.
- Human-Presence Radar (24 GHz FMCW Human-Presence Radar – HLK-LD2410B): This is the last sensor used in the project's design, and it has fixed every problem that occurred throughout the research. It is installed under the table and is distinguished by its excellent capacity to identify both fixed items and human presence.

2. Performance and real-time response limitations: Data must be processed and sent in real-time in order to update the library's indicators and ensure correct information display without delay.

3. Cybersecurity and privacy Limitations: These include avoiding the use of cameras that respect users' privacy, reducing psychological stress, anxiety, and negative emotions, safeguarding devices and the internet network from possible risks to digital security, and protect the system from any potential.

3.2.5 Design Standards

To ensure effective communication, device security, and ease of use, the system is based on a collection of international standards and protocols:

IEEE 802.11:(WIFI):

IEEE 802.11 is the standard used for wireless local area networks (WLANs). In the proposed system, the ESP32 uses Wi-Fi based on IEEE 802.11 to connect to the network and transmit data wirelessly to the server.[31]

OASIS MQTT IoT communication (Message Queuing Telemetry Transport):

For large-scale specialised message exchange in Internet of Things (IoT) applications, MQTT is a simple protocol. The ESP32 main processor unit in the suggested system uses MQTT to send data that has been measured by a number of sensors. Because of its low bandwidth and power usage, this protocol is suitable for devices with limited resources.[34]

IEEE C95.1 electromagnetic safety: Electromagnetic Safety Standard, IEEE C95.1-2019:

This standard sets safety limits to protect people from negative health consequences caused by electromagnetic field radiation in the frequency range of 0 hertz to 300 gigahertz. This standard acts as a manual for the safe operation of the wireless communications system and the 24 GHz HLK-LD2410B mmWave Radar sensor in the suggested system.

3.2.6 Design Alternatives

Design Aspect	Alternative Option	Limitations	Selected Solution	Why Selected
MicrocontrollerUnit	Raspberry Pi 4	High cost bulky size need high power overly complex	ESP32 Microcontroller	Low power Low cost fully optimized for embedded IoT tasks. integrated Wi-Fi supports all required communication protocols.
Occupancy Sensors	PIR Sensors Optical Cameras	PIR sensors fail to detect stationary occupants; optical cameras raise severe privacy concerns for students.	mmWaveRadar (HLK-LD2410B)	Detection of both moving and stationary people , Ability to conceal it within a non-metallic housing , It includes a delay time adjustment feature
Communication Protocol	HTTP	High Overhead High Power Consumption High Latency Poor Efficiency on Weak Networks	MQTT Protocol	Lightweight publish-subscribe architecture, low bandwidth usage, and ideal for constrained IoT devices

Table 1 : 3.2.6 Design Alternatives

3.2.7 Safety Considerations

Due to the nature of the system, which is made up of electronic devices and sensors placed in a public setting like a library, it is necessary to pay attention to safety standards and put rules and regulations in place to protect staff and library users from potential risks as well as to protect user privacy and data security.

Among these standards are the following:

1. User Safety: Visitors' and employees' safety will be the top priority. To reduce any potential electrical risks, the system's design will rely on electronic components and control circuits that operate at low voltage and with a high degree of safety.

The central processing unit and the many sensors will be carefully hidden beneath the tables inside special, hermetically sealed protective enclosures to ensure both environmental and physical safety. In order to prevent any accidental contact, lower the risk of electric shock, and lessen the possibility of visitors messing with the components, this procedure will completely insulate all electrical connections and wires and keep them out of people's direct reach. This will provide a stable and safe learning environment for everyone.

2. Work Team Safety During Implementation:

Future stages of development and field installation will take into account the safety of the work team and those involved in putting the project concept into effect.Close respect to general safety regulations is planned for the design and implementation of the electrical circuits.To ensure a safe learning environment throughout the project's implementation period, these precautions will include using electrical insulation tools and safe soldering equipment, Be careful when handling connections, and following safety rules when installing the main processing unit inside the library.

3. User privacy, data security, and safety:

In addition to making sure that users and employees are safe, extra care will be taken to secure data that is sent and received via the system. Specifically, the transmission of sensor readings within the system will be set up in a way that prevents any access without permission or external breaches. Additionally, by avoiding the usage of monitoring cameras, the solution will protect user privacy.



Figure 14 : 3.2.7 Sealed Protective
[10]

3.2.8 Design considerations table

Table 2 : 3.2.8 Design Elements of the Suggested System

Design consideration	Project application	Relevant location in report
Performance	Real-time data processing and transmission, and dashboard updates.	Sec3.2.4&Sec3.2.2
serviceability	The components will be designed for easy installation, maintenance, and fault diagnosis.	Sec3.2.1&Sec3.2.2
Economic	Low-cost, energy-efficient components such as the ESP32 were selected.	Sec3.2.1&Sec3.2.6
Environmental	A system has been designed to suit general conditions and monitor temperature and noise levels.	Sec3.2.1&Sec3.2.2
Environmental Sustainability	The selected components are considered to reduce energy consumption.	Sec2.4&Sec3.2.1
Manufacturability	The system uses compact, simple sensor units that are easy to install.	Sec3.2.1
Ethical	The system prohibits the use of surveillance cameras to protect user privacy and avoid the psychological impact of surveillance.	Sec2.4&Sec3.2.3&Sec3.2.7
Health and safety	Low-voltage electronic components were selected, and sensors, wiring, and control units were protected within the main electrical enclosures; electromagnetic safety will also be taken into account.	Sec3.2.5&Sec3.2.7
Social	Administrative workers can efficiently organise and manage projects due to the system's clarity and specialised area designed to meet the specific needs of schools. With a simple online interface tailored to their individual needs, special attention has been given to ensuring accessibility for all user groups.	Sec2.3&Sec3.2.1
Political	For the suggested system, no specific political variables were found. However, institutional requirements and applicable telecommunications regulations are considered.	Sec3.2.3&Sec3.2.5

4.1 Summarize the main contributions of the work.

Finding an environment that works for their study style is one of the most difficult tasks for users of public libraries, especially those at universities. Overcrowded spaces and disorganisation often result in lost time and elevated stress, which affects users' focus and attention; also, the conventional techniques of randomly looking for a seat have become impossible. Users' problems will be reduced by being provided with an intelligent, integrated system for keeping watch on, setting up, and managing library spaces—without the need for any methods that would risk the privacy of others. The system's design is built on the combination of artificial intelligence (AI) and Internet of Things (IoT) technology to improve their combined efficiency; it has several sensors that collaborate to measure different data points and present them on an accessible dashboard . In order to ensure their inclusion and give everyone a comfortable and equal experience, the system also gives special attention to users with specific requirements.Our technology will significantly improve the lives of users and assist staff members in managing resources with little expense and effort, resulting in a welcoming and accessible learning environment for everyone.

4.2 Further future work someone should do to make the solution/system better.

Although the proposed system provides the basic functions required for managing a smart library, there is room for future enhancements that would make the user experience even smoother. The following are the most notable future enhancements:

1. Integration with the university's website

2. Full integration with the Internet of Things

3. Suggest a seating area near books related to the user's field of study.

4. Recommend books that align with the user's academic major or interests.

5. Encourage users not to sit for long periods by providing reminders or suggesting short breaks for movement.

6. Provide an audio version of a library glossary or terminology list for people with visual impairments.

7. Suggest a seating area near books related to the user's field of study.

8. A sensor measures the ambient temperature, and based on the readings, the air conditioning system is automatically controlled.

9. When noise levels exceed the specified threshold, the system sends a warning message to those inside the room.

10. Add a sensor to detect fires and smoke and send alert notifications to those present.

11. the possibility to use digital twin technology to extend the library system to smart cities and academic campuses.

4.3 Lessons learned

While working on this project, we gained a great deal of experience and technical skills, the most important and notable of which are:

1. We learned that choosing a sensor depends on several factors.
2. We learned the differences between various sensors and how each one works.
3. We learned that the initial design may require several revisions before arriving at the appropriate design.
4. We learned the importance of teamwork, time management, and task delegation.
5. We learned the importance of environmental conservation by selecting eco-friendly components.
6. We developed our skills in overcoming the various and unexpected challenges we faced.
7. We expanded our knowledge in the field of the Internet of Things (IoT) and electronic components.

Reference

[1]

https://www.sciencedirect.com/science/article/abs/pii/S0099133315000555

[2]

https://www.researchgate.net/publication/355721628_IoT-Based_Smart_Library_Seat_Occupancy_and_Reservation_System_using_RFID_and_FSR_Technologies_for_South_African_Universities_of_Technology

[3]

https://pmc.ncbi.nlm.nih.gov/articles/PMC9029361/

[4]

https://www.sciencedirect.com/science/article/abs/pii/S0099133312001206

[5]

https://www.sciencedirect.com/science/article/abs/pii/S009913330900069X

[6]

https://www.researchgate.net/profile/Gordon-Ononiwu/publication/280558358_RADIO_FREQUENCY_IDENTIFICATION_RFID_BASED_ATTENDANCE_SYSTEM_WITH_AUTOMATIC_DOOR_UNIT/links/55b9176a08ae9289a08ff882/RADIO-FREQUENCY-IDENTIFICATION-RFID-BASED-ATTENDANCE-SYSTEM-WITH-AUTOMATIC-DOOR-UNIT.pdf

[7]

https://iris.polito.it/handle/11583/2703783

[8]

https://www.intechopen.com/chapters/88506

[9]

https://www.researchgate.net/publication/413525534_Librarians_Application_of_Internet_of_Things_IoT_in_Libraries

[10]

https://www.researchgate.net/publication/374448728_Internet_of_Things_IOT_and_Libraries

[11]

https://www.sciencedirect.com/topics/computer-science/artificial-intelligence

[12]

https://www.sciencedirect.com/science/article/pii/S2542660525002938

[13]

https://www.mdpi.com/2079-9292/12/24/4925

[14]

https://www.sciencedirect.com/science/article/abs/pii/S0378778823003687

[15]

https://www.sciencedirect.com/science/article/abs/pii/S0099133320300343

[16]

https://www.tandfonline.com/doi/full/10.1080/13614533.2017.1414066

[17]

http://www.ijcee.org/show-36-244-1.html

[18]

https://learn.adafruit.com/force-sensitive-resistor-fsr

[19]

https://www.tekscan.com/blog/flexiforce/how-does-force-sensing-resistor-fsr-work

[20]

https://www.sciencedirect.com/science/article/abs/pii/S0924424720317416

[21]

https://www.ti.com/lit/an/snoa927a/snoa927a.pdf?ts=1791208242163&ref_url=https%253A%252F%252Fwww.google.com%252F

[22]

https://www.researchgate.net/publication/361818927_Airfoil_Anemometer_With_Integrated_Flexible_Piezo-Capacitive_Pressure_Sensor/figures

[23]

https://www.mdpi.com/2078-2489/12/2/87?utm_source=chatgpt.com#Most_Common_IoT_Architectures

[24]

https://docs.aws.amazon.com/wellarchitected/latest/modern-industrial-data-technology-lens/midaperf01-bp02..html

[25]

https://www.espressif.com/en/products/socs/esp32

[26]

https://h.hlktech.com/download/HLK-LD2410B-24G/1/HLK%20LD2410B%E7%94%9F%E5%91%BD%E5%AD%98%E5%9C%A8%E6%84%9F%E5%BA%94%E6%A8%A1%E7%BB%84%E8%AF%B4%E6%98%8E%E4%B9%A6%20V1.0%209.pdf

[27]

https://components101.com/modules/inmp441-mems-omnidirectional-microphone#:~:text=INMP441%20MEMS%20High%20Precision%20Omnidirectional%20Microphone%20Module.,for%20advanced%20audio%20and%20voice%20recognition%20applications.

[28]

https://eur-lex.europa.eu/EN/legal-content/summary/general-data-protection-regulation-gdpr.html

[29]

https://www.mouser.do/en/new/hammond-manufacturing/hammond-rz-water-tight-enclosures/

[30]

https://www.icj-e.org/download/ICJE-8-6-149-158.pdf

[31]

https://www.researchgate.net/publication/358343601_Characterization_of_IEEE80211_WLAN_Implementing_on_Excavator_Can_We_Use_IEEE80211_for_i-construction

[32]

https://ink.library.smu.edu.sg/cgi/viewcontent.cgi?article=4308&context=sis_research

[33]

https://www.icj-e.org/download/ICJE-8-6-149-158.pdf

[34]

https://www.researchgate.net/publication/406849756_Detecting_anomalies_in_MQTTMQTT-SN_traffic_using_intelligent_learning_models

https://trepo.tuni.fi/bitstream/handle/10024/161838/978-952-03-3707-0.pdf

https://www.tekscan.com/blog/flexiforce/how-does-force-sensing-resistor-fsr-work

https://www.researchgate.net/publication/361818927_Airfoil_Anemometer_With_Integrated_Flexible_Piezo-Capacitive_Pressure_Sensor/figures

Figure References

[1]

https://trepo.tuni.fi/bitstream/handle/10024/161838/978-952-03-3707-0.pdf

MFRC-522 RC522 RFID Radiofrequency IC Card Inducing Sensor Reader for Arduino | eBay

[2]

https://www.tekscan.com/blog/flexiforce/how-does-force-sensing-resistor-fsr-work

[3]

https://www.researchgate.net/publication/361818927_Airfoil_Anemometer_With_Integrated_Flexible_Piezo-Capacitive_Pressure_Sensor/figures

[4]

https://h.hlktech.com/download/HLK-LD2410B-24G/1/HLK%20LD2410B%E7%94%9F%E5%91%BD%E5%AD%98%E5%9C%A8%E6%84%9F%E5%BA%94%E6%A8%A1%E7%BB%84%E8%AF%B4%E6%98%8E%E4%B9%A6%20V1.0%209.pdf

[5]

https://community.element14.com/technologies/embedded/b/blog/posts/experimenting-with-microwave-based-sensors-for-presence-detection

[6]

https://manuals.plus/hi-link/hlk-ld2410b-radar-module-manual.pdf

[7]

https://zaitronics.com.au/products/inmp441-i2s-mems-omnidirectional-microphone

[8]

https://learn.adafruit.com/assets/26680

[9]

https://www.researchgate.net/publication/351245647_Cross-Layer_Design_in_Internet_of_Things_IOT-Issues_and_Possible_Solutions/figures?lo=1

[10]

https://www.mouser.do/en/new/hammond-manufacturing/hammond-rz-water-tight-enclosures/